

Graph Neural Networks versus Tabular Machine Learning for Illicit Transaction Detection in Bitcoin: An Empirical Evaluation under Chronological Partitioning

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Abstract—The detection of illicit transactions in public cryptocurrency ledgers is a critical challenge in financial forensic technology and regulatory compliance. Graph Neural Networks (GNNs) have emerged as an intuitive paradigm for blockchain analysis, operating on the premise that topological message passing over transaction networks will outperform conventional machine learning models. However, past empirical benchmarks frequently employ random cross-validation or feature-rich baselines that obscure the isolated inductive benefit of graph structure. In this work, we present a rigorous, reproducible empirical comparison between three foundational Graph Neural Network architectures (GCN, GraphSAGE, GAT) and four strong conventional machine learning baselines (Logistic Regression, Random Forest, XGBoost, MLP) using the Elliptic Bitcoin benchmark under a strict chronological evaluation protocol ($T = 49$ time steps). Our experimental design controls for feature representations (93 local features vs. 165 full features), class-imbalance loss formulations (standard vs. cost-sensitive weighted cross-entropy), and multi-seed parameter initialization ($N = 5$ random seeds). Across the full 165-dimensional feature space, conventional tree-based ensembles demonstrate strong performance, with Random Forest achieving a holdout test illicit F1-score of 0.7247 ± 0.0027 (PR-AUC 0.6599 ± 0.0025) and XGBoost achieving 0.7131 ± 0.0111 (PR-AUC 0.6744 ± 0.0028), compared to 0.3308 ± 0.0602 for GAT, 0.2822 ± 0.1248 for GraphSAGE, and 0.2589 ± 0.0639 for GCN. We observe that tabular models benefit consistently from pre-engineered 1-hop aggregate features ($\Delta F1 = +0.0311$ to $+0.0430$), whereas GNN sensitivity to feature representation is architecture-dependent. Furthermore, extensive ablation studies reveal that GNN performance is highly sensitive to graph directionality, unknown-node topological context, and random initialization seed variance (GCN F1 range 0.4686; GraphSAGE F1 range 0.3669). Statistical comparisons between the validation-selected GNN (GraphSAGE on local features, test F1 0.4153 ± 0.1914) and tree baselines yield raw paired t -test differences ($p \approx 0.023$) that do not remain statistically significant following Holm-Bonferroni correction ($p_{\text{Holm}} = 0.0917$) or exact Wilcoxon signed-rank testing ($p = 0.0625$), bounded by small sample power limits ($N = 5$). All models exhibit substantial temporal generalization degradation (relative F1 drops of 23.6% to 44.9%) across chronological splits, driven by marked non-stationarity in illicit transaction prevalence. Our findings demonstrate that spatial graph convolution does not automatically outperform well-tuned tabular tree ensembles on

financial transaction data, establishing baseline recommendations for empirical blockchain forensics.

Index Terms—Bitcoin, illicit transaction detection, fraud detection, graph neural networks, GraphSAGE, GCN, GAT, tabular machine learning, temporal generalization, class imbalance.

I. Introduction

A. Background and Context

Decentralized blockchain protocols, most prominently Bitcoin, have introduced trustless, pseudonymous financial transaction networks [1], [2]. While providing unprecedented financial autonomy and cryptographic settlement guarantees, public ledgers have also been exploited by illicit actors for ransomware extortion, money laundering, darknet contraband trafficking, and sanctions evasion [1], [3].

In response, regulatory bodies and cryptocurrency analytics firms have invested heavily in automated blockchain forensic systems to identify anomalous and illicit transactional behavior [4], [5]. Forensic analysis is made possible by the public availability of the Bitcoin transaction graph, where addresses and transactions form a directed, time-stamped acyclic graph of value transfers [2], [6].

B. Problem Definition and Class Imbalance

The automated identification of fraudulent entities in cryptocurrency transaction graphs represents a specialized supervised node classification problem under severe label skew and temporal non-stationarity. In practice, confirmed illicit transactions constitute a minute fraction of total transaction volume ($< 2\%$), while legitimate commercial transfers, exchange arbitrage, and mining payouts dominate the network [3], [7]. Furthermore, in public benchmarks such as the Elliptic Bitcoin dataset [8], over 77% of transactions lack deanonymized ground-truth labels.

Forensic classification systems must therefore generalize under three compounding constraints:

- 1) Extreme Class Imbalance: A high ratio of licit to illicit entities ($\approx 9.25 : 1$ among labeled nodes, and $> 40 : 1$ globally) [7].
- 2) High Structural Sparsity and Unlabeled Context: A massive majority (77.15%) of graph nodes remain unannotated, forcing algorithms to propagate structural information across unlabeled paths [8].
- 3) Temporal Non-Stationarity and Adversarial Concept Drift: Laundering typologies, coin-mixing strategies, and illicit operations evolve dynamically over time, leading to severe distribution shift across chronological epochs [9], [10].

C. Motivation for Graph Representation Learning

Because Bitcoin transactions are natively interconnected through directed payment edges, Graph Neural Networks (GNNs) [11], [12] have emerged as a natural methodological paradigm for blockchain forensics. Unlike conventional tabular machine learning architectures that evaluate transactions as isolated feature vectors, GNNs leverage spatial message passing to recursively aggregate representations from multi-hop topological neighborhoods [13]–[15].

In theory, neighborhood aggregation allows GNNs to exploit relational patterns that are inaccessible to tabular models, such as fan-out mixing patterns, pass-through intermediary nodes, and multi-hop fund consolidation [8].

D. Research Gap and Primary Contributions

Despite theoretical motivation, prior studies comparing GNNs and tabular baselines on blockchain data have often exhibited methodological limitations:

- 1) Confounded Feature Spaces: Tabular models are frequently compared using raw metadata, while GNNs are provided with pre-aggregated topological features, conflating graph convolution with feature engineering [16], [17].
- 2) Random Split Contamination: Standard randomized train/test splits allow information from future time steps to leak into training partitions, violating real-world causality [18], [19].
- 3) Inadequate Ablation of Graph Construction: The isolated contributions of edge directionality, unlabeled node inclusion, and layer depth are rarely dissected systematically under identical optimization constraints.

To address these gaps, this paper presents a controlled empirical evaluation of Graph Neural Networks and tabular machine learning under strict chronological partitioning on the Elliptic Bitcoin benchmark. Our primary contributions are:

- Strict Chronological Benchmarking: We evaluate four tabular baselines (Logistic Regression, Random Forest, XGBoost, MLP) and three GNN architectures (GCN, GraphSAGE, GAT) across 49 discrete time steps under a prospective train/validation/test split.

- Isolated Feature Representation Analysis: We independently evaluate performance on intrinsic local metadata (93 features) versus the full feature space (165 features) containing 1-hop topological aggregates.
- Systematic Topological Ablations: We quantify the impact of unlabeled graph context, edge directionality (forward, backward, bidirectional), message-passing depth (1–3 layers), and cost-sensitive loss weighting.
- Multi-Seed Statistical and Error Characterization: We report performance distributions across $N = 5$ seeds, conduct paired hypothesis tests with Holm-Bonferroni correction, and provide degree-resolved error taxonomies.

E. Paper Organization

The remainder of this manuscript is organized as follows: Section II formalizes the dataset, feature representation, and chronological evaluation protocol. Section III details the baseline and GNN architectures, loss formulations, and statistical methodology. Section IV presents the main benchmark results. Section V details topological and architectural ablation studies. Section VI provides formal statistical significance and power analyses. Section VII presents error taxonomies and graph-structural characterization. Section VIII discusses scientific and practical implications. Section IX outlines methodological limitations, and Section X concludes with directions for future work.

II. Dataset and Problem Formulation

A. The Elliptic Bitcoin Benchmark

We conduct our empirical investigation on the Elliptic Bitcoin transaction dataset [8], comprising $|V| = 203,769$ transaction nodes and $|E| = 234,355$ directed payment edges partitioned across $T = 49$ discrete, chronologically ordered time steps. Each time step represents an approximately two-week window of blockchain activity. All directed edges exist strictly between transactions within the same time step ($t_u = t_v$ for all $(u, v) \in E$); there are zero cross-timestep edges [8].

B. Entity Classes and Label Imbalance

Transactions are classified into three mutually exclusive categories:

- Class 1 (Illicit): Verified malicious entities, including darknet marketplaces, ransomware extortionists, scams, and mixing services ($N = 4,545$, 2.23% total, 9.76% labeled).
- Class 2 (Licit): Verified legitimate entities, including exchanges, wallet services, miners, and legal merchants ($N = 42,019$, 20.62% total, 90.24% labeled).
- Class "unknown" (Unlabeled): Transactions lacking ground-truth attribution ($N = 157,205$, 77.15% total).

The ratio of licit to illicit transactions among labeled nodes is approximately 9.245 : 1 (see Fig. 1).

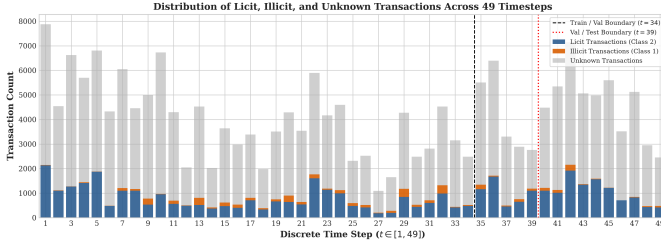


Fig. 1. Distribution of Licit, Illicit, and Unknown transactions across the 49 discrete timesteps in the Elliptic benchmark dataset.

C. Evaluation Metrics under Class Skew

Given severe class imbalance, overall accuracy is uninformative [20], [21]. We evaluate minority-class detection using:

$$\text{Precision (P)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1)$$

$$\text{Recall (R)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{F1-Score} = \frac{2 \cdot \text{P} \cdot \text{R}}{\text{P} + \text{R}} \quad (3)$$

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (4)$$

Our primary target metric is the Illicit-Class F1-Score on the blind test holdout ($t \in [40, 49]$), supported by Precision-Recall Area Under the Curve (PR-AUC).

D. Feature Space Representation

Each transaction node $v \in V$ possesses a 165-dimensional feature vector $x_v \in \mathbb{R}^{165}$ consisting of:

- 1) Local Features (1–93, $d_{\text{local}} = 93$): Intrinsic meta-data, including transaction fees, transacted volumes, input/output counts, and script properties.
- 2) Aggregated Features (94–165, $d_{\text{agg}} = 72$): Hand-crafted 1-hop summary statistics (minimum, maximum, mean, standard deviation) computed across immediate predecessor and successor transactions [8].

E. Chronological Partitioning Protocol

To evaluate prospective temporal generalization, we enforce a strict chronological split (see Fig. 2 and Table I):

- Training Set ($t \in [1, 34]$): 29,894 labeled nodes (3,462 illicit, 26,432 licit; 11.58% illicit prevalence).
- Validation Set ($t \in [35, 39]$): 5,486 labeled nodes (447 illicit, 5,039 licit; 8.15% illicit prevalence).
- Test Set ($t \in [40, 49]$): 11,184 labeled nodes (636 illicit, 10,548 licit; 5.69% illicit prevalence).

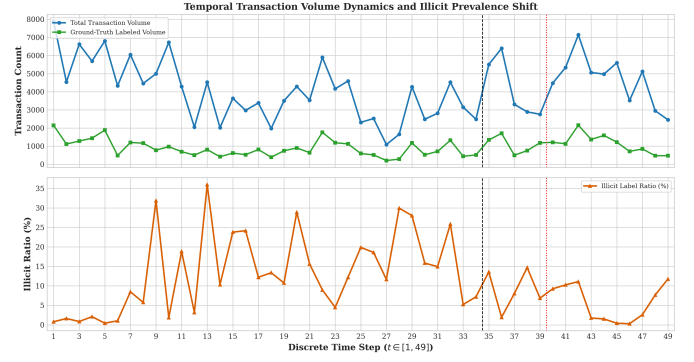


Fig. 2. Total transaction volume and temporal class composition over the 49 discrete timesteps, showing chronological training, validation, and test partitions.

III. Methodology

A. Baseline Model Architectures

We benchmark against four conventional machine learning baselines:

- Logistic Regression (LR): Linear classification optimized via L-BFGS with L_2 regularization ($C = 1.0$).
- Random Forest (RF): Bagged ensemble of 100 decision trees ($n_{\text{estimators}}=100$, $\text{max_depth}=\text{None}$, $\text{min_samples_split}=2$) [22].
- XGBoost: Gradient-boosted decision trees ($n_{\text{estimators}}=100$, $\text{max_depth}=6$, $\text{learning_rate}=0.1$, $\text{subsample}=0.8$, $\text{colsample_bytree}=0.8$) [23].
- Multi-Layer Perceptron (MLP): Deep neural network with 2 hidden layers (128 units each), ReLU activations, Dropout ($p = 0.3$), Batch Normalization, trained via Adam ($lr = 0.001$, weight decay 10^{-5}).

B. Graph Neural Network Architectures

We evaluate three foundational Graph Neural Network architectures:

- Graph Convolutional Networks (GCN) [13]:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right) \quad (5)$$

- GraphSAGE [14]:

$$h_v^{(l+1)} = \sigma \left(W^{(l)} \cdot \left[h_v^{(l)} \parallel \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} h_u^{(l)} \right] \right) \quad (6)$$

- Graph Attention Networks (GAT) [15]:

$$\alpha_{vu} = \frac{\exp(\text{LeakyReLU}(a^T [Wh_v \parallel Wh_u]))}{\sum_{k \in \mathcal{N}(v)} \exp(\text{LeakyReLU}(a^T [Wh_v \parallel Wh_k]))} \quad (7)$$

All GNN models employ 2 message-passing layers, 128 hidden channels, Dropout ($p = 0.3$), and Adam optimizer ($lr = 0.001$). GAT utilizes 4 attention heads (32 channels per head). Hyperparameter details are provided in Table II.

TABLE I
Elliptic Dataset Structure and Temporal Split Summary

Metric / Attribute	Training Partition	Validation Partition	Test Partition	Full Dataset
Timestep Range (t)	Timesteps 1–34	Timesteps 35–39	Timesteps 40–49	Timesteps 1–49
Discrete Time Steps	34 snapshots	5 snapshots	10 snapshots	49 snapshots
Labeled Transactions	29,894 (64.18%)	5,486 (11.78%)	11,184 (24.04%)	46,564 (100.0%)
Licit Transactions	26,432 (88.42%)	5,039 (91.85%)	10,548 (94.31%)	42,019 (90.24%)
Illicit Transactions	3,462 (11.58%)	447 (8.15%)	636 (5.69%)	4,545 (9.76%)
Imbalance Ratio (Licit : Illicit)	7.63 : 1	11.27 : 1	16.58 : 1	9.245 : 1
Total Graph Nodes (incl. Unknown)	136,005	23,790	43,974	203,769
Total Graph Edges	156,589	27,243	50,523	234,355

TABLE II
Model Architecture and Hyperparameter Specification

Model	Family	Key Hyperparameters
Logistic Reg.	Linear	$C = 1.0$, L_2 penalty, L-BFGS solver
Random Forest	Bagged Trees	100 trees, unlimited depth, Gini split
XGBoost	Boosting	100 trees, depth 6, lr 0.1, sub 0.8
MLP	Neural Net	2 layers, 128 units, Dropout 0.3, Adam
GCN	Graph Conv	2 layers, 128 hidden, Adam (10^{-3})
GraphSAGE	Inductive GNN	2 layers, 128 hidden, Mean aggregate
GAT	Attention GNN	2 layers, 128 hidden, 4 heads (32/head)

C. Loss Formulations and Cost-Sensitive Weighting

We evaluate unweighted cross-entropy alongside cost-sensitive class weighting ($w_{\text{pos}} = 26,432/3,462 \approx 7.6349$). Unknown-label nodes are retained in the graph for message passing but strictly masked out from supervised loss computation:

$$\mathcal{L}_{\text{masked}} = -\frac{1}{|\mathcal{V}_{\text{train}}^{\text{labeled}}|} \sum_{v \in \mathcal{V}_{\text{train}}^{\text{labeled}}} [w_{\text{pos}} y_v \log \hat{y}_v + (1 - y_v) \log(1 - \hat{y}_v)] \quad (8)$$

D. Training and Threshold Locking Protocol

Models are trained for 100 epochs on Timesteps 1–34 with Early Stopping (patience = 15) monitoring validation illicit F1 on Timesteps 35–39. Decision thresholds are tuned strictly on the validation partition ($\tau^* = \arg \max F1_{\text{val}}$) and locked prior to blind evaluation on Timesteps 40–49.

E. Statistical Significance Testing Protocol

All non-deterministic architectures are evaluated across $N = 5$ random seeds ($S = \{42, 123, 456, 789, 999\}$). Statistical comparisons between the validation-selected GNN and baselines employ two-sided paired t -tests with Holm-Bonferroni correction [24] and exact paired Wilcoxon signed-rank tests [25], [26].

IV. Results

A. Main Comparison on Full Features

Quantitative results on the full 165-feature representation across five seeds are reported in Table III and illustrated in Fig. 3:

- Random Forest (Standard): Achieved test illicit F1 of 0.7247 ± 0.0027 and PR-AUC of 0.6599 ± 0.0025 .
- XGBoost (Standard): Achieved test illicit F1 of 0.7131 ± 0.0111 and PR-AUC of 0.6744 ± 0.0028 .
- MLP (Standard): Achieved test illicit F1 of 0.5848 ± 0.0194 and PR-AUC of 0.5115 ± 0.0187 .
- Logistic Regression (Standard): Produced deterministic test illicit F1 of 0.4015 and PR-AUC of 0.2754.
- GAT (Weighted): Produced the highest full-feature GNN performance (F1 0.3308 ± 0.0602 , PR-AUC 0.2667 ± 0.0695).
- GraphSAGE (Weighted): Produced F1 of 0.2822 ± 0.1248 and PR-AUC of 0.2366 ± 0.1386 .
- GCN (Weighted): Produced F1 of 0.2589 ± 0.0639 and PR-AUC of 0.2202 ± 0.0703 .

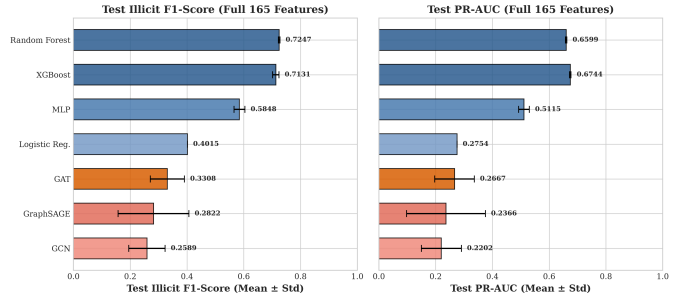


Fig. 3. Main benchmark comparison of test illicit F1-score and PR-AUC between conventional tabular machine learning baselines and Graph Neural Network architectures on the 165-feature representation.

B. F1 and PR-AUC Dynamics

For XGBoost, continuous ranking quality was superior (PR-AUC 0.6744) despite Random Forest obtaining a marginally higher discrete F1 (0.7247 vs. 0.7131). Across GNNs, test F1 and PR-AUC tracked closely together (0.3308 vs. 0.2667 for GAT; 0.2822 vs. 0.2366 for GraphSAGE; 0.2589 vs. 0.2202 for GCN).

TABLE III
Main Benchmark Results on Full Features (165 Dimensions)

Model Architecture	Model Family	Feat.	Loss Form	Validation F1	Test Illicit F1	Test PR-AUC	Test Prec. / Rec.
Random Forest	Tree Ensemble	165	Standard	0.9489 ± 0.0023	0.7247 ± 0.0027	0.6599 ± 0.0025	0.9586/0.5827
XGBoost	Gradient Boosting	165	Standard	0.9399 ± 0.0053	0.7131 ± 0.0111	0.6744 ± 0.0028	0.9127/0.5855
MLP	Neural Network	165	Standard	0.8212 ± 0.0344	0.5848 ± 0.0194	0.5115 ± 0.0187	0.7630/0.4761
Logistic Regression	Linear Model	165	Standard	0.5921 ± 0.0000	0.4015 ± 0.0000	0.2754 ± 0.0000	0.3821/0.4230
GAT	Graph Attention	165	Weighted	0.6006 ± 0.0982	0.3308 ± 0.0602	0.2667 ± 0.0695	0.2680/0.4412
GraphSAGE	Inductive GNN	165	Weighted	0.4473 ± 0.1590	0.2822 ± 0.1248	0.2366 ± 0.1386	0.2695/0.3588
GCN	Graph Conv Net	165	Weighted	0.4293 ± 0.0923	0.2589 ± 0.0639	0.2202 ± 0.0703	0.2788/0.2884

C. Precision-Recall Trade-offs

Precision-Recall dynamics (see Fig. 4) reveal that tree ensembles achieved exceptionally high precision (0.9586 for RF; 0.9127 for XGBoost) at moderate recall (≈ 0.58), whereas GNN architectures produced lower precision (0.2680–0.2788) and moderate recall (0.2884–0.4412).

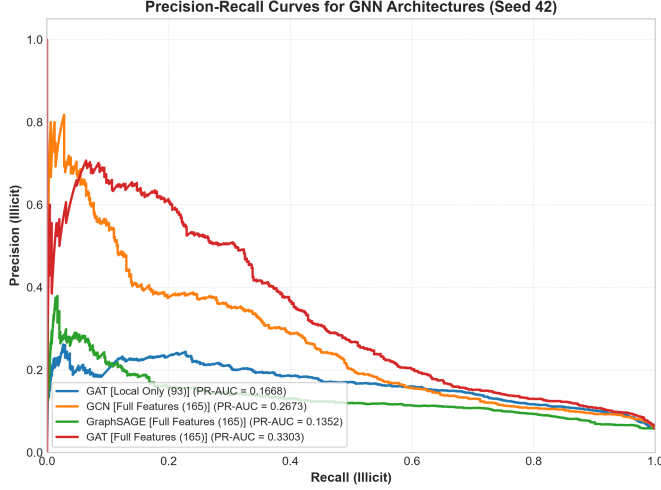


Fig. 4. Precision-Recall curves for primary Graph Neural Network architectures evaluated on the blind test holdout partition across timesteps 40–49.

D. Local versus Full Feature Representations

Table IV and Fig. 5 compare models trained on Local 93 features versus Full 165 features:

- Tree ensembles and MLP benefited consistently from 1-hop aggregate features (RF: +0.0311; XGB: +0.0430; MLP: +0.0187).
- GAT test F1 improved by +0.0942 (0.2367 $\rightarrow$ 0.3308).
- In contrast, GraphSAGE achieved its highest performance on 93 Local features (0.4153 ± 0.1914), declining by -0.1331 on Full features (0.2822 ± 0.1248).
- GCN was largely invariant ($\Delta F1 = +0.0030$).

E. Standard versus Cost-Sensitive Loss Weighting

As shown in Table V, positive-class loss weighting ($w_{\text{pos}} \approx 7.63$) shifted precision-recall operating points rather than providing universal F1 gains, as validation

TABLE IV
Feature Representation Comparison (Local 93 vs. Full 165)

Model	Local 93 F1	Full 165 F1	$\Delta F1$	$\Delta \text{PR-AUC}$
Random Forest	0.6936 ± 0.0022	0.7247 ± 0.0027	+0.0311	+0.0267
XGBoost	0.6701 ± 0.0069	0.7131 ± 0.0111	+0.0430	+0.0355
MLP	0.5661 ± 0.0232	0.5848 ± 0.0194	+0.0187	+0.0227
Logistic Reg.	0.4378 ± 0.0000	0.4015 ± 0.0000	-0.0363	-0.0252
GAT	0.2367 ± 0.0825	0.3308 ± 0.0602	+0.0942	+0.1014
GraphSAGE	0.4153 ± 0.1914	0.2822 ± 0.1248	-0.1331	-0.1293
GCN	0.2559 ± 0.0768	0.2589 ± 0.0639	+0.0030	+0.0064

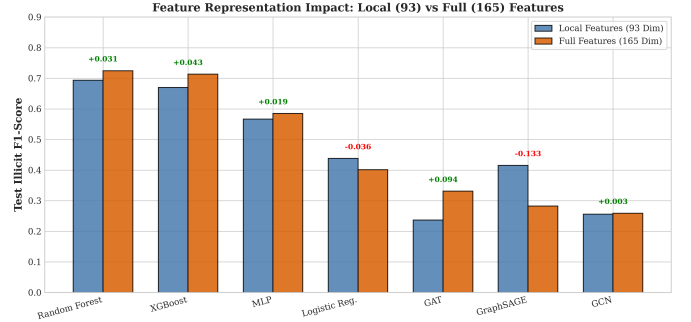


Fig. 5. Feature representation impact comparing Local 93 metadata features against Full 165 features across conventional ML and GNN models.

threshold tuning had already recalibrated unweighted models.

TABLE V
Class Loss Weighting Comparison (Full 165 Features)

Model Architecture	Standard F1	Weighted F1	$\Delta F1$	MCC (Std)
Random Forest	0.7247 ± 0.0027	0.7024 ± 0.0051	-0.0223	0.7367 ± 0.0057
XGBoost	0.7131 ± 0.0111	0.7197 ± 0.0074	+0.0066	0.7189 ± 0.0154
MLP	0.5848 ± 0.0194	0.5513 ± 0.0362	-0.0334	0.5840 ± 0.0155
Logistic Reg.	0.4015 ± 0.0000	0.3480 ± 0.0000	-0.0535	0.3640 ± 0.0000

F. Validation-to-Test Generalization Decay

All architectures suffered marked performance declines from validation (Timesteps 35–39) to blind test (Timesteps 40–49):

$$\Delta_{\text{rel}} = \frac{\text{Val F1} - \text{Test F1}}{\text{Val F1}} \times 100 \quad (9)$$

Relative drops: RF: -23.6% ; XGBoost: -24.1% ; MLP: -28.8% ; GraphSAGE (Local): -31.5% ; GCN: -39.7% ; GAT: -44.9% .

G. Random-Seed Stability and Initialization Variance

Table VI and Fig. 9 demonstrate that tree ensembles exhibited tight cross-seed clustering ($\sigma \leq 0.0111$), while GNNs displayed wide dispersion across seeds (GraphSAGE Local range $\Delta = 0.3669$, GCN Full range $\Delta = 0.4686$).

TABLE VI
Multi-Seed Stability and Parameter Initialization Sensitivity

Model Configuration	Min F1	Max F1	Range (Δ)	Mean $\pm$ Std F1
Random Forest (165)	0.7209	0.7271	0.0062	0.7247 ± 0.0027
XGBoost (165)	0.6984	0.7223	0.0239	0.7131 ± 0.0111
MLP (165)	0.5520	0.5989	0.0469	0.5848 ± 0.0194
Logistic Reg. (165)	0.4015	0.4015	0.0000	0.4015 ± 0.0000
GAT (165)	0.2241	0.3638	0.1397	0.3308 ± 0.0602
GraphSAGE (165)	0.1477	0.4746	0.3269	0.2822 ± 0.1248
GCN (165)	0.1723	0.6409	0.4686	0.2589 ± 0.0639
GraphSAGE (Local 93)	0.2583	0.6252	0.3669	0.4153 ± 0.1914

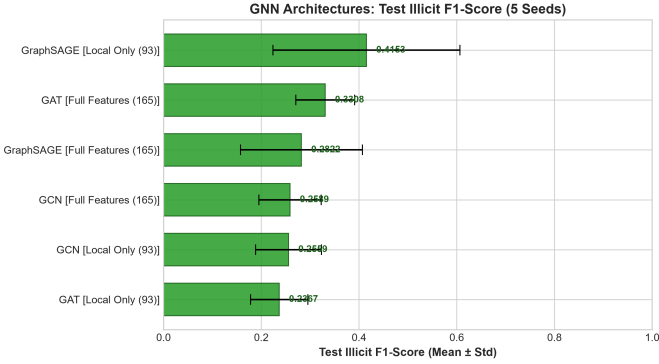


Fig. 6. Cross-seed test F1 score distributions and parameter initialization variance across five random seeds.

V. Ablation and Sensitivity Analysis

A. Impact of Unlabeled Graph Context

Table VII and Fig. 6 present the ablation comparing the full 100% topology against an induced subgraph of labeled nodes only:

- For GCN, removing unknown nodes increased test illicit F1 from 0.2589 to 0.4555 ($\Delta = +0.1966$).
- For GraphSAGE, removing unknown nodes increased F1 from 0.2822 to 0.4132 ($\Delta = +0.1310$).
- For GAT, retaining unknown nodes produced higher F1 (0.3308 vs. 0.2918).

TABLE VII
Unknown-Node Context Ablation (Timesteps 40–49, 165 Features)

Architecture	Retained F1	Removed F1	Δ F1	Trend
GAT	0.3308 ± 0.0602	0.2918 ± 0.0521	-0.0390	Retained Higher
GCN	0.2589 ± 0.0639	0.4555 ± 0.0524	$+0.1966$	Removed Higher
GraphSAGE	0.2822 ± 0.1248	0.4132 ± 0.0514	$+0.1310$	Removed Higher

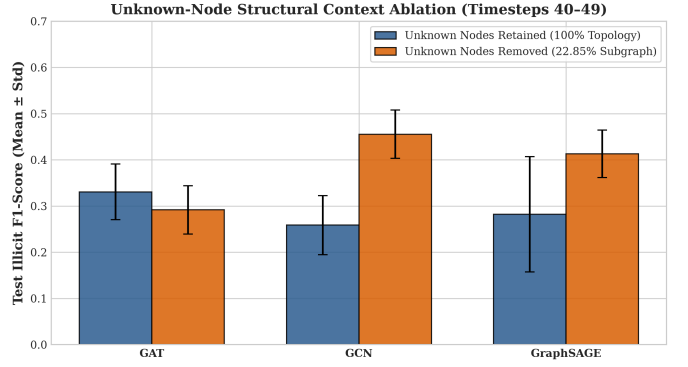


Fig. 7. Unknown-node context ablation comparing test F1 performance when unlabeled transaction nodes are retained versus removed from the graph.

B. Graph Propagation Directionality

Table VIII and Fig. 7 examine Forward ($u \rightarrow v$), Backward ($v \rightarrow u$), and Bidirectional message passing. Backward propagation yielded higher retrospective test F1 (0.3213–0.3953) than forward flow (0.2589–0.3308), and bidirectional propagation reached 0.4618 for GAT. However, backward flow relies on future transactional context unavailable during live broadcast; forward flow remains the primary inductive standard.

TABLE VIII
Graph Directionality Ablation (Timesteps 40–49, 165 Features)

Architecture	Forward	Backward	Bidirectional	Δ (Bwd-Fwd)
GAT	0.3308 ± 0.0602	0.3938 ± 0.0706	0.4618 ± 0.0234	$+0.0630$
GCN	0.2589 ± 0.0639	0.3953 ± 0.0462	0.3142 ± 0.0571	$+0.1364$
GraphSAGE	0.2822 ± 0.1248	0.3213 ± 0.0558	0.3710 ± 0.1228	$+0.0391$

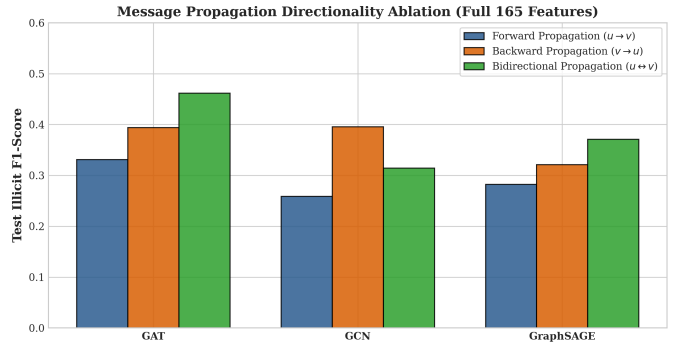


Fig. 8. Graph directionality ablation comparing Forward, Backward, and Bidirectional message-passing propagation across GNN architectures.

C. Message-Passing Layer Depth

Table IX and Fig. 8 show depth responses across 1, 2, and 3 layers:

- GCN degraded monotonically ($0.2862 \rightarrow 0.2589 \rightarrow 0.2089$), consistent with theoretical over-smoothing in isotropic convolutions [27]–[29].

- GAT peaked at 2 layers (0.3308 ± 0.0602).
- GraphSAGE achieved its highest full-feature performance with a single layer (0.3750 ± 0.1215).

TABLE IX
GNN Depth Ablation (Timesteps 40–49, 165 Features)

Architecture	1 Layer	2 Layers	3 Layers	Profile
GAT	0.2877 ± 0.0319	0.3308 ± 0.0602	0.2847 ± 0.0052	Peak at 2
GCN	0.2862 ± 0.0250	0.2589 ± 0.0639	0.2089 ± 0.0892	Monotonic
GraphSAGE	0.3750 ± 0.1215	0.2822 ± 0.1248	0.3327 ± 0.1250	Peak at 1

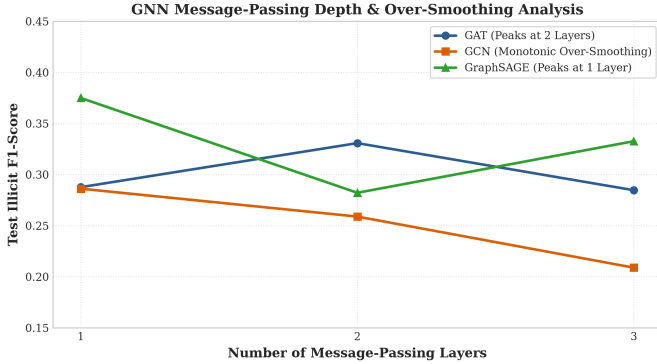


Fig. 9. Message-passing depth ablation across 1, 2, and 3 layers for GCN, GraphSAGE, and GAT.

VI. Statistical Analysis

A. Paired Hypothesis Testing

To eliminate post-hoc selection bias, we evaluate the validation-selected GNN (GraphSAGE on Local 93 Features, Val F1 = 0.6060, Test F1 = 0.4153 ± 0.1914 , Test PR-AUC = 0.3659 ± 0.1859) against representative baselines across five shared seeds ($N = 5$). As reported in Table X:

- Unadjusted paired t -tests indicated differences favoring Random Forest ($p_{\text{raw}} = 0.0229$, $d_z = -1.6062$) and XGBoost ($p_{\text{raw}} = 0.0247$, $d_z = -1.5694$) [30], [31].
- Following Holm-Bonferroni correction [24], adjusted p -values increased to $p_{\text{Holm}} = 0.0917$.
- Exact paired Wilcoxon signed-rank tests produced $p_{\text{Wilcoxon}} = 0.0625$ [25], [26].
- Consequently, differences do not cross the formal significance threshold of $\alpha = 0.05$ after multiple comparison adjustment.

B. Statistical Power Bounds at $N = 5$

For sample size $N = 5$, there are $2^5 = 32$ possible signed-rank permutations. The minimum achievable two-sided Wilcoxon p -value is mathematically bounded at:

$$p_{\min} = \frac{2}{2^5} = \frac{2}{32} = 0.0625 \quad (10)$$

It is therefore mathematically impossible for a two-sided Wilcoxon test with $N = 5$ to achieve $p < 0.05$, regardless of effect magnitude [25], [26].

VII. Error and Graph-Structural Analysis

A. Error Taxonomy and Degree Disparity

Table XI details error distributions and connectivity across the 11,184 test nodes (402 TP, 8,165 TN, 2,383 FP, 234 FN):

- True Negatives had the highest connectivity (mean in-degree 2.38, out-degree 1.34).
- True Positives had the lowest connectivity (mean in-degree 0.87, out-degree 0.70).
- The test set contained zero high-confidence false positives ($\hat{p} \geq 0.80$) and zero high-confidence false negatives ($\hat{p} \leq 0.20$), indicating that misclassifications occurred in regions of model uncertainty.

B. Graph Homophily and Relational Topology

Of 234,355 directed edges, only 36,624 (15.63%) connect two labeled nodes. Among mutually labeled edges, same-class homophily is 95.37% (34,928/36,624):

$$h_{\text{labeled}} = \frac{34,928}{36,624} = 95.37\% \quad (11)$$

Conversely, 197,731 edges (84.37%) involve at least one unknown node (see Fig. 10).

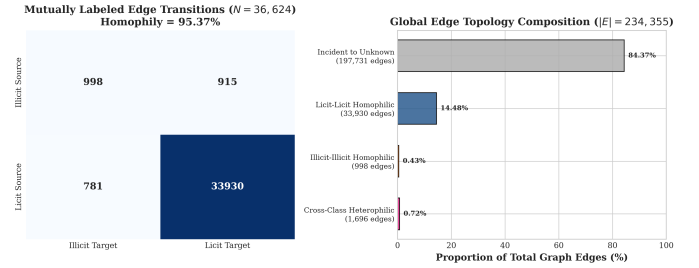


Fig. 10. Edge class transition matrix and relational homophily distribution among labeled and unlabeled transaction nodes.

C. Temporal Error Patterns across Timesteps 40–49

Analyzing individual test timesteps (see Fig. 11) reveals that during Timesteps 40–42, illicit prevalence was high (9.25%–11.10%, test F1 up to 0.5234). At Timestep 43, illicit transaction prevalence collapsed by 84% (11.10% → 1.75%), driving test F1 down to 0.0718.

VIII. Discussion

A. Comparative Performance of GNNs versus Tabular Baselines

Under the evaluated chronological protocol, tree-based ensembles (Random Forest F1 0.7247, XGBoost F1 0.7131) outperformed the evaluated GNN models (GAT 0.3308, GraphSAGE 0.2822, GCN 0.2589) on full features. This outcome is conditional on the benchmark setup and does not constitute universal proof of GNN inferiority [16], [18].

TABLE X
Paired Statistical Significance Tests (GraphSAGE Local 93 vs. Baselines, $N = 5$ Seeds)

Metric / Comparison	Mean Diff (Δ)	Paired t	p (Raw)	p (Holm)	W -Stat	p (Wilcoxon)	Cohen d_z	FWER Signif.
Test Illicit F1								
vs. Random Forest	-0.3094	-3.5917	0.0229	0.0917	0.0	0.0625	-1.6062	No ($p \geq 0.05$)
vs. XGBoost	-0.3044	-3.5092	0.0247	0.0917	0.0	0.0625	-1.5694	No ($p \geq 0.05$)
vs. MLP	-0.1695	-1.8872	0.1322	0.2644	3.0	0.3125	-0.8440	No ($p \geq 0.05$)
vs. Logistic Regression	+0.0673	+0.7856	0.4760	0.4760	6.0	0.8125	+0.3513	No ($p \geq 0.05$)
Test PR-AUC								
vs. Random Forest	-0.2940	-3.5141	0.0246	0.0816	0.0	0.0625	-1.5716	No ($p \geq 0.05$)
vs. XGBoost	-0.3064	-3.7246	0.0204	0.0816	0.0	0.0625	-1.6657	No ($p \geq 0.05$)
vs. MLP	-0.1456	-1.6440	0.1755	0.3070	3.0	0.3125	-0.7352	No ($p \geq 0.05$)
vs. Logistic Regression	+0.1462	+1.7584	0.1535	0.3070	3.0	0.3125	+0.7864	No ($p \geq 0.05$)

TABLE XI
Error and Graph-Structural Taxonomy (Test Set, $N = 11,184$)

Category / Segment	Count	Prop. (%)	In-Deg.	Out-Deg.
Total Test Nodes	11,184	100.00%	2.06	1.26
True Positives (TP)	402	3.59%	0.87	0.70
True Negatives (TN)	8,165	73.01%	2.38	1.34
False Positives (FP)	2,383	21.31%	1.23	1.13
False Negatives (FN)	234	2.09%	1.61	0.87
High-Confidence FP ($\hat{p} \geq 0.80$)	0	0.00%	0.00	0.00
High-Confidence FN ($\hat{p} \leq 0.20$)	0	0.00%	0.00	0.00

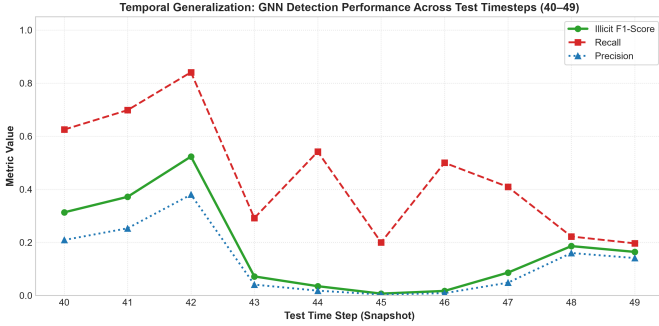


Fig. 11. Timestep-by-timestep test performance breakdown across timesteps 40–49, illustrating performance collapse during the Timestep 43 regime shift.

B. Critical Evaluation of Relational Inductive Bias

The availability of relational transaction structure does not guarantee that spatial GNNs will outperform strong tabular models [11], [32]. When tabular classifiers receive 1-hop pre-engineered aggregates, decision trees partition feature space orthogonally without propagating raw feature noise across sparse neighborhoods [16], [22], [23].

C. Feature Representation Dynamics

Handcrafted 1-hop aggregates improved Random Forest (+0.0311 F1), XGBoost (+0.0430 F1), MLP (+0.0187 F1), and GAT (+0.0942 F1), but degraded GraphSAGE (-0.1331 F1), indicating architecture-specific feature interactions.

D. Class Imbalance and Threshold Tuning Dynamics

Because models underwent validation threshold tuning ($\tau^* = \arg \max F1_{\text{val}}$), unweighted models adjusted decision thresholds effectively. Loss weighting ($w_{\text{pos}} = 7.63$) shifted precision-recall operating trade-offs without offering consistent F1 gains.

E. Temporal Distribution Shift and Concept Drift

All architectures experienced substantial validation-to-test drops (23.6% to 44.9%), reflecting temporal distribution shift [9], [10].

F. Seed Sensitivity and Optimization Landscape

Tree models showed tight cross-seed clustering ($\sigma \leq 0.0111$), while GNNs displayed substantial initialization dispersion (GraphSAGE Local range $\Delta = 0.3669$, GCN Full range $\Delta = 0.4686$).

G. Structural Role of Unlabeled Transactions

GCN and GraphSAGE achieved higher F1 on the isolated labeled subgraph (0.2589 $\rightarrow$ 0.4555 and 0.2822 $\rightarrow$ 0.4132), whereas GAT performed best when unknown nodes were retained (0.2918 $\rightarrow$ 0.3308).

H. Causal Directionality versus Retrospective Leakage

Backward propagation yielded higher retrospective test F1 (0.3213–0.3953), but relies on future context unavailable during live transaction broadcast; forward flow remains the proper inductive standard.

I. Over-Smoothing and Layer Depth Dynamics

GCN degraded monotonically with depth (0.2862 $\rightarrow$ 0.2589 $\rightarrow$ 0.2089), consistent with potential over-smoothing across isotropic convolutions [27]–[29]. GAT peaked at 2 layers (0.3308), and GraphSAGE peaked at 1 layer (0.3750).

J. Statistical Power and Multiple Comparison Corrections

While raw paired t -tests indicated differences favoring tree ensembles ($p_{\text{raw}} \approx 0.023$), adjusted hypothesis tests did not cross $\alpha = 0.05$ after Holm-Bonferroni correction ($p_{\text{Holm}} = 0.0917$) or Wilcoxon signed-rank testing ($p = 0.0625$) due to small sample power bounds ($N = 5$) [24]–[26].

K. Structural Profiles of Model Misclassifications

True positives exhibited lower connectivity (mean degree < 1.0) than true negatives (mean degree > 2.0), indicating that isolated illicit nodes present unique forensic detection challenges.

L. Darknet Market Dynamics and Timestep 43 Non-Stationarity

The sharp drop in illicit transaction volume at Timestep 43 underscores the vulnerability of static forensic models to sudden macroeconomic and law enforcement interventions [8], [19].

M. Practical Implications for Industry and Regulators

In production compliance environments, well-tuned gradient boosting ensembles with engineered topological aggregates offer strong detection capability, low inference latency, and high parameter stability.

IX. Limitations

A. Anonymized Feature Semantics

Features in the Elliptic benchmark are fully anonymized, precluding domain-specific feature interpretation.

B. High Proportion of Unlabeled Transactions

77.15% of transactions lack labels, limiting full relational ground truth.

C. Temporal Non-Stationarity and Regime Shifts

Adversarial adaptation and darknet market disruptions introduce substantial concept drift [9], [10].

D. Statistical Power Constraints with Five Seeds

Evaluating $N = 5$ random seeds limits the mathematical power of non-parametric statistical tests ($p_{\min} = 0.0625$).

E. Static Architecture Limitations

Evaluated GNNs operate on static graph snapshots rather than continuous-time dynamic graphs.

F. Homogeneous Graph Assumptions

Transactions and payment flows are modeled as homogeneous graphs, omitting heterogeneous entity types [19].

G. Dataset-Specific Generalization Boundaries

Conclusions are established on the Elliptic Bitcoin dataset and require validation on other blockchain networks.

H. Discrete Temporal Snapshots

Transactions are aggregated into 2-week snapshots rather than continuous transaction streams.

X. Conclusion and Future Work

A. Conclusion

This study presented a controlled empirical comparison of Graph Neural Networks and conventional tabular machine learning on the Elliptic Bitcoin dataset under chronological partitioning. Tree ensembles (Random Forest and XGBoost) demonstrated strong performance on the full feature representation, benefiting consistently from 1-hop topological aggregates. In contrast, GNN performance proved highly sensitive to graph directionality, unlabeled context, layer depth, and parameter initialization. Spatial graph convolution does not automatically guarantee superiority over well-engineered tabular models in financial transaction forensics.

B. Future Work

Future research should explore continuous-time dynamic graph neural networks [33], [34], heterogeneous transaction graphs, and drift-adaptive online learning mechanisms.

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TABLE XII
Research Question Answer Matrix

RQ ID	Core Investigation	Quantitative Finding	Principal Scientific Conclusion
RQ1	GNN vs. Tabular Performance	RF F1: 0.7247, XGB F1: 0.7131 vs. GAT: 0.3308, SAGE: 0.2822, GCN: 0.2589	Tree ensembles outperformed GNNs under the evaluated setup; graph convolution did not automatically surpass tabular models.
RQ2	Feature Space & Loss Formulations	1-Hop Features: +0.03 to +0.04 for trees, +0.09 for GAT, -0.13 for SAGE	Aggregated features benefit tree models consistently, while GNN responses are architecture-dependent.
RQ3	Topological Context, Direction & Depth	Unknowns: GCN/SAGE +0.13–+0.20 without unknowns; GAT +0.04 with unknowns	GNN sensitivity to unlabeled context, directionality, and depth is architecture-specific.
RQ4	Temporal Shift, Seed Variance & Errors	Val-to-Test drop: -23.6% to -44.9%; GNN seed range up to 0.4686	Severe temporal non-stationarity and initialization sensitivity impact model reliability.

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